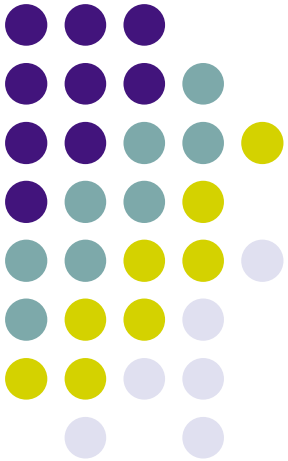




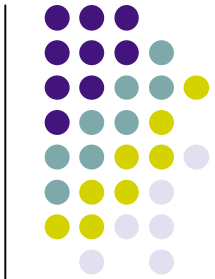
ASP.NET on Unix

with mod_mono



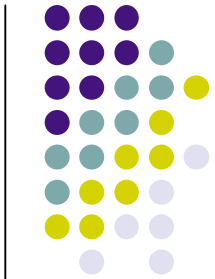
Daniel López Ridruejo
daniel@rawbyte.com

About me



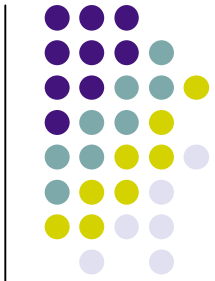
- Open source: Author of mod_mono, Comanche, several Linux Howtos and the Teach Yourself Apache 2 book
- Company: founder of BitRock, multiplatform installers and management software

About this presentation



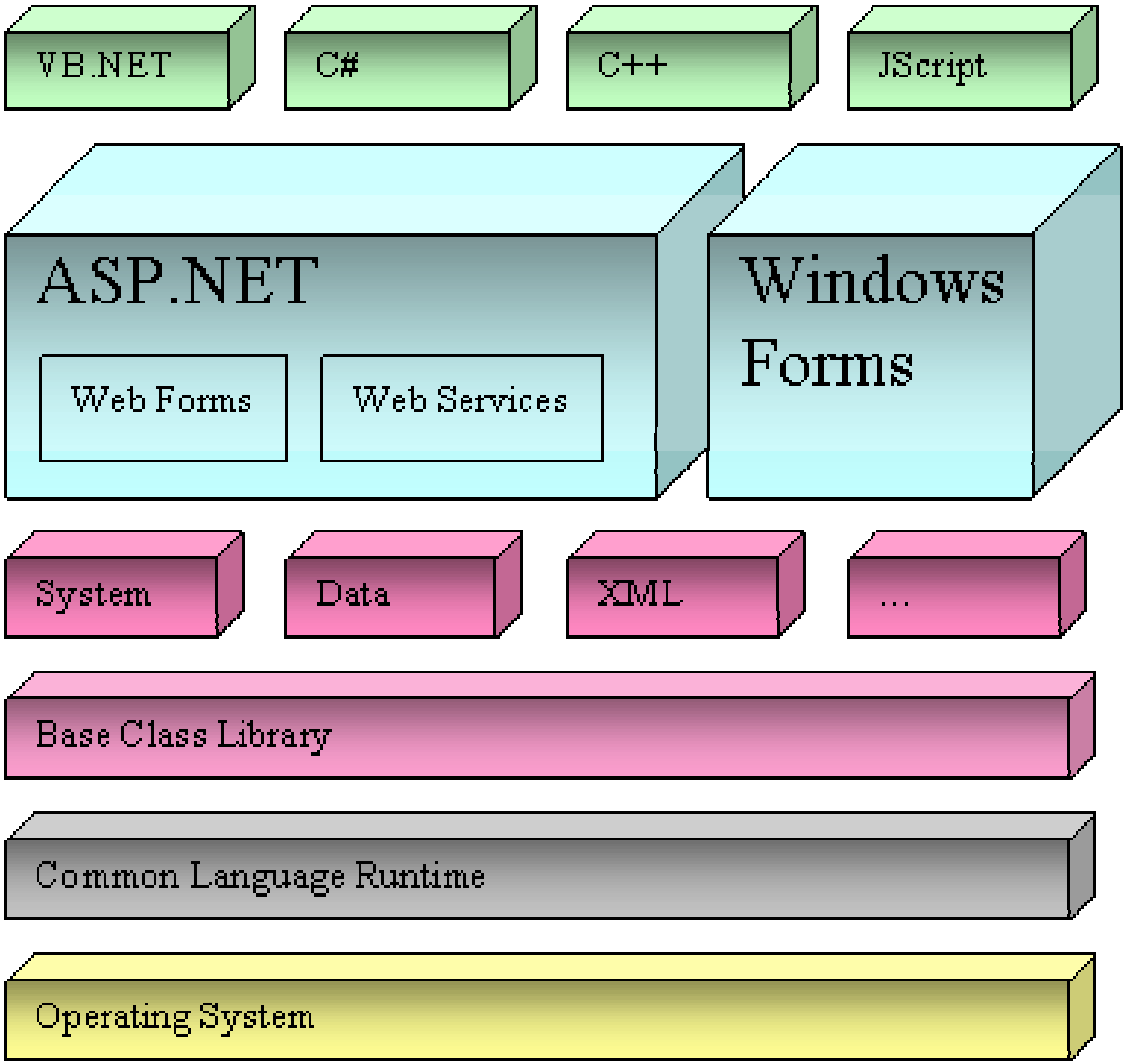
- The .NET framework
- An overview of Mono, a multiplatform implementation of the .NET framework
- mod_mono : run ASP.NET on Linux using Apache and Mono

The Microsoft .Net initiative

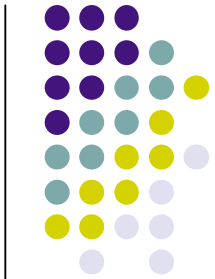


- Many things to many people, we are interested in a subset: the .NET framework
- Common Language Runtime execution environment
- Comprehensive set of class libraries
- As other technologies, greatly hyped. But is a solid technical foundation that stands on its own merits

The .NET Framework

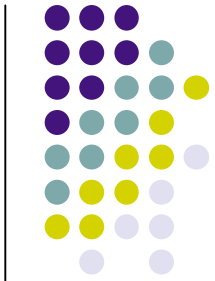


.NET Highlights (1)



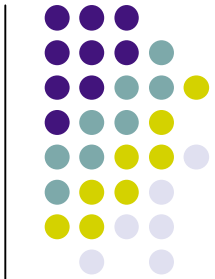
- Common Language Runtime : Provides garbage collection, resource management, threads, JIT support and so on. Somewhat similar to JVM
- Common Intermediate Language, multiple language support: C#, Java, Visual Basic, C++, JavaScript, Perl, Python, Eiffel, Fortran, Scheme, Pascal Cobol… They can interoperate and have access to full .NET capabilities. Similar to Java bytecode

.NET Highlights (2)



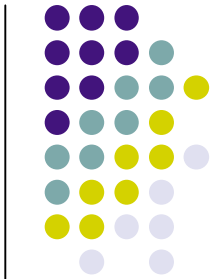
- Comprehensive Class Library: XML, ASP.NET, Windows Forms, Web Services, ADO.NET
- Microsoft wants .NET to succeed in other platforms: standardization effort at ECMA, Rotor sample implementation.
- P/Invoke: Easy to incorporate and interoperate with existing code
- Attributes, delegates, XML everywhere

Mono Project



- Open Source .NET framework implementation. Covers ECMA standard plus additional technologies: Windows Forms, ASP.NET, ADO.NET
- Started by Ximian, looking for better development tools for Linux desktop
- Incredible progress in short amount of time, hundreds of contributors

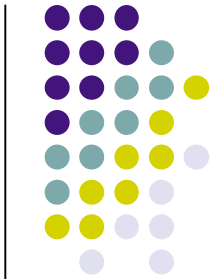
What does Mono provide?



- C# compiler
- Runtime
- Class libraries
- Tools: debugger, documentation browser

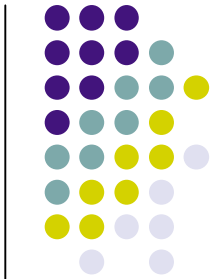
Supports Linux, FreeBSD, Windows, Mac OS X, IBM mainframes and many more on the works

Related Projects



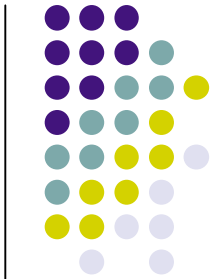
- GUI development:
  - System.Windows.Forms (Wine/GTK)
  - Gtk#
  - QT#
  - SharpWT
- JScript, Visual Basic bundled with Mono, other third party, IKVM project
- SharpDevelop IDE

Why Mono for Linux development?



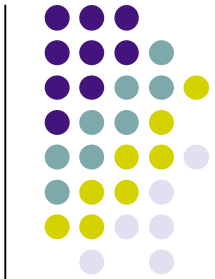
- Ease desktop development
- Reuse of existing code libraries, extend applications.
- Ease Windows migration. Develop on Visual Studio, deploy on Linux.
- Multiple language support and seamless interoperability

ASP.NET



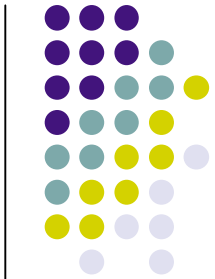
- Server side development with .NET languages, mainly C# and Visual Basic.
- Component based development
- xcopy deployment
- Data binding
- Two approaches:
  - Traditional: code intermixed with HTML
  - Code behind

Why ASP.NET?



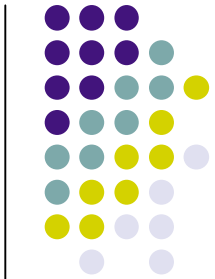
- Unified approach to server side development
  - Languages
  - Tools
  - Libraries
- Simplified development and deployment
  - Event model
  - Configuration
  - Shadow copying

Http Pipeline



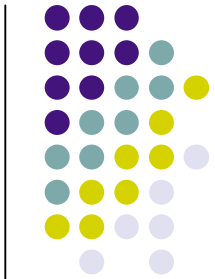
- Similar to Apache hook / filter system
- Create a handler or a module and register interest in events during the life of the request (BeginRequest, AuthenticateRequest, ResolveRequestCache, and so on)
- Important classes: HttpApplication, HttpWorkerRequest, HttpRequest

Caching



- Output Caching: caching full page, fragments
- Possible and easy to implement custom caching modules
- Data caching engine available to developers

State management



- Application
- Session : HttpSession
  - In memory
  - State server
  - Database
- Cookie
- View

Demo Time!



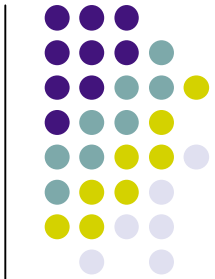
Welcome to Mono XSP!



Here are some ASP.NET examples:

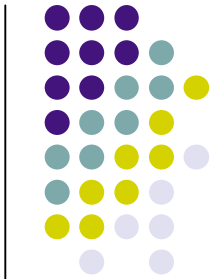
- button.aspx
- calendar.aspx
- codebehind1.aspx
- code-render.aspx
- databind-arraylist.aspx
- databind-attribute.aspx
- databind-class.aspx
- databind-template.aspx
- dbpage1.aspx

mod_mono



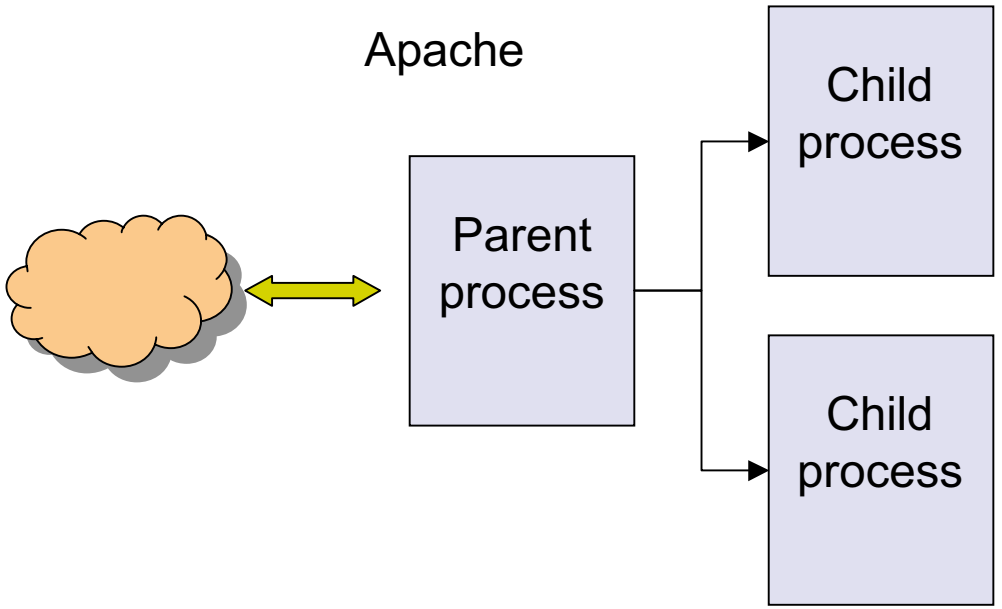
- Allows running ASP.NET on Linux using Apache and Mono
- Apache handles the HTTP protocol, creates HttpWorkerRequest and passes it to the ASP.NET pipeline for processing
- ASP.NET processes the request and produces a result that Apache returns to the client

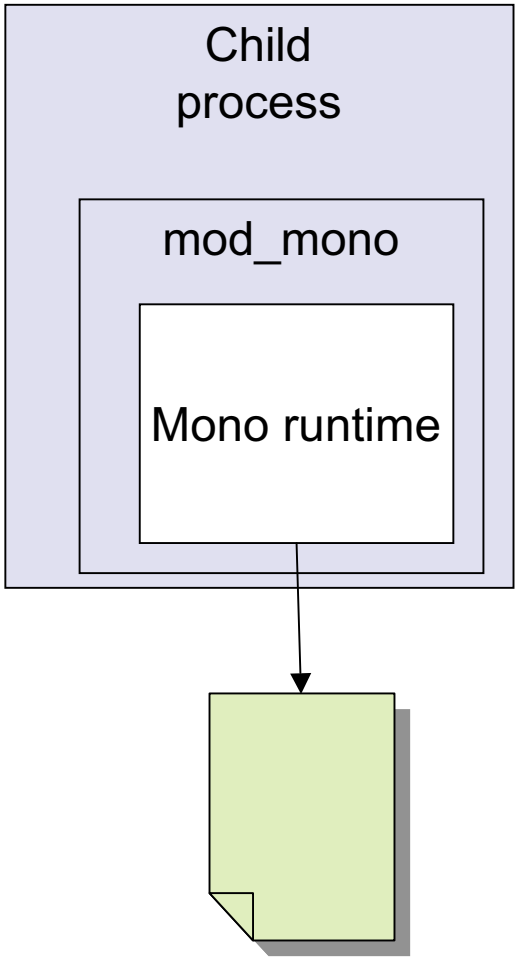
mod_mono features



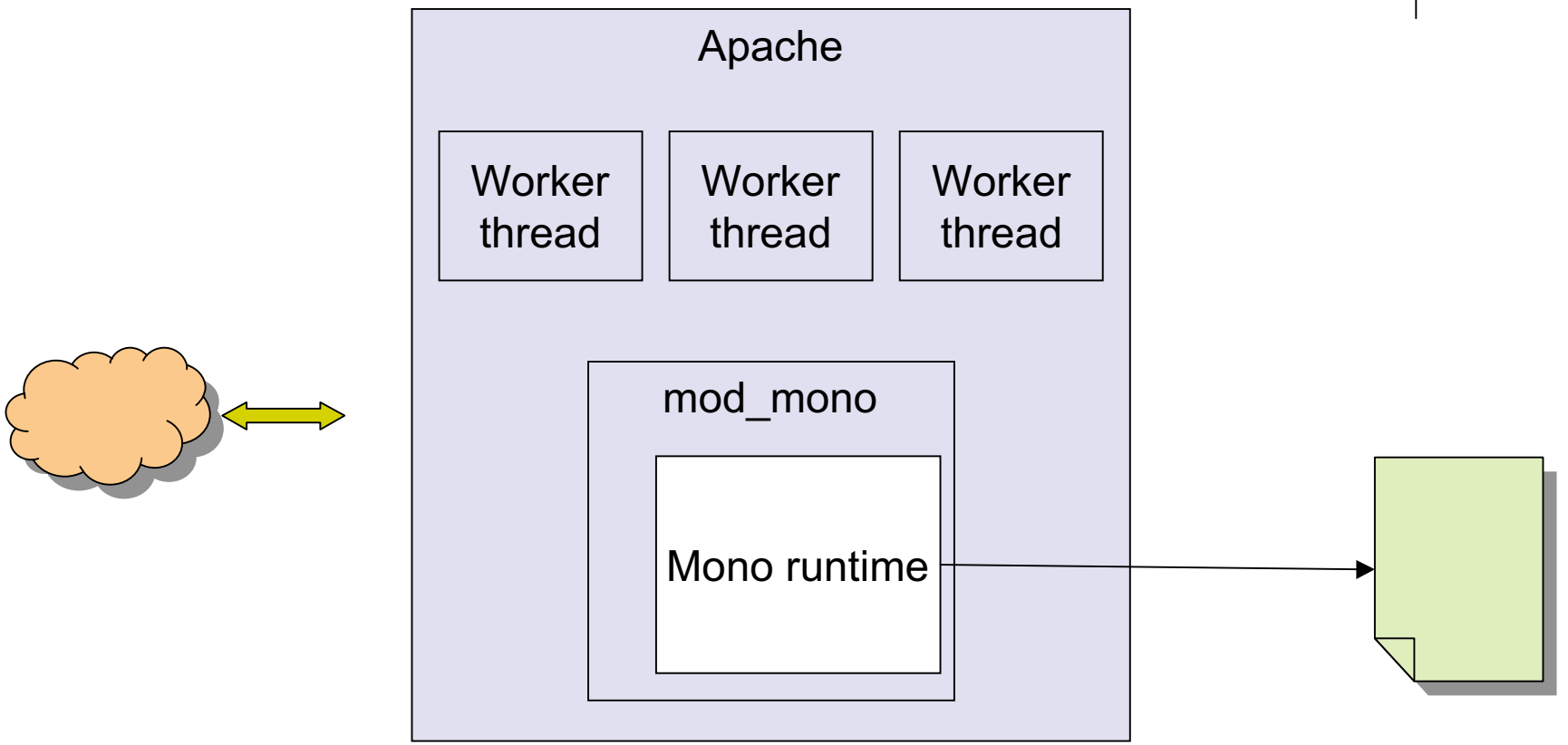
- Supports Apache 2 and 1.3 versions
- Portable, should run on any platform that Apache and Mono support, including Windows. Current support Linux and FreeBSD
- Processing model, tradeoff reliability, session support, scalability, performance
  - Multiple processes
  - Purely threaded
  - Out of process

Multiple processes

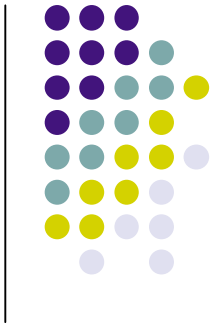


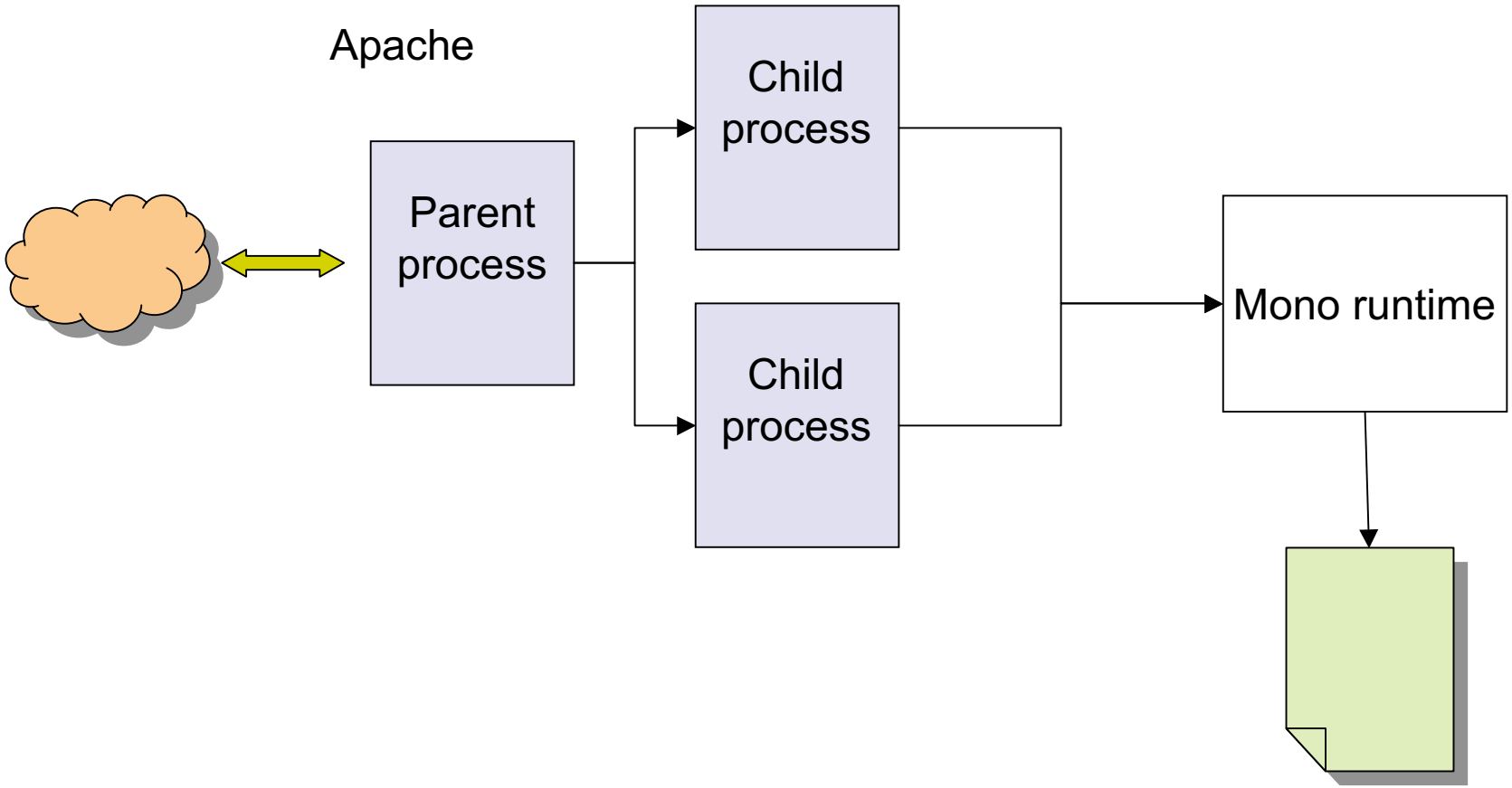


Threaded Apache

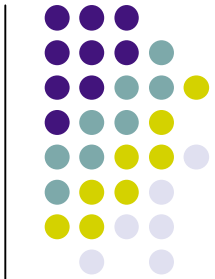


Out of process



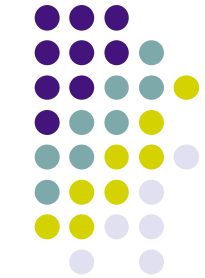


Mono Misconceptions



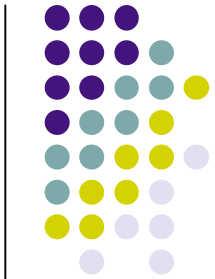
- Always playing catch up
- Patent enforcement
- C# / Java
- License issues

Advantages of Mono ASP.NET for Enterprise environments



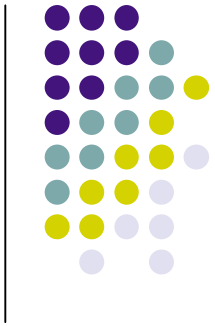
- Reuse current Windows expertise, code
- Component based development
- Develop on Windows, deploy on Linux/Unix
  - Windows have better development tools
  - Linux regarded as more secure, scalable, reliable
  - Avoid license fees
- Interoperate with other open source technologies
- Reduce costs and prevent vendor lock-in, providing a migration path

The future



- Mono
  - Version 1.0 planned for early next year
  - Support for .NET 1.1 features such as generics
- mod_mono
  - Windows support
  - Additional processing models
- Get involved!

Q&A



Thanks for Attending!

More info:
http://www.go-mono.com
http://www.apacheworld.org/modmono/

You can reach me at:
daniel@rawbyte.com

This presenation has ended.
Click anywhere to return.

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